

Portfolio Analysis and Optimisation of a Diversified U.S. Equity Portfolio

An Application of Modern Portfolio Theory

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Abstract

This report presents a quantitative analysis of a diversified portfolio of 30 U.S. equities spanning seven sectors, technology, financials, healthcare, consumer goods, automobiles, energy and industrials, over the five-year period from January 2020 to January 2025. Historical closing price data obtained from Yahoo Finance were used to compute daily and cumulative returns, from which an equally weighted portfolio, each constituent assigned a weight of 3.33%, was constructed and evaluated using established techniques from Modern Portfolio Theory.

The equal-weighted portfolio produced an expected annual return of 23.78%, an annualised volatility of 21.40% and a Sharpe Ratio of 0.92, with a maximum drawdown of 33.24% recorded during the COVID-19 market shock of early 2020. A Monte Carlo simulation of 10,000 randomly weighted portfolios was conducted to approximate the Efficient Frontier, identifying a Maximum Sharpe Portfolio (29.53% return, 22.60% volatility, Sharpe Ratio of 1.13) and a Minimum Volatility Portfolio (20.44% return, 18.55% volatility, Sharpe Ratio of 0.89).

The results demonstrate that portfolio allocation, rather than asset selection alone, is a principal determinant of risk-adjusted performance. Reweighting the portfolio toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon improved efficiency relative to the equal-weight benchmark without a proportional increase in risk. The report concludes with a discussion of the practical implications of these findings for portfolio construction and the limitations of Modern Portfolio Theory in real-world markets.

Keywords: Modern Portfolio Theory, portfolio optimisation, Monte Carlo simulation, Efficient Frontier, Sharpe Ratio, diversification, risk-adjusted return

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1. Introduction

1.1 Background

Investors constructing a portfolio of risky assets face a fundamental trade-off between expected return and risk. Modern Portfolio Theory, introduced by Harry Markowitz, provides the analytical framework through which this trade-off can be quantified and optimised, and remains the theoretical foundation for the majority of contemporary portfolio construction techniques. This report applies that framework to a diversified portfolio of U.S. equities, using a complete Python-based analytical workflow that progresses from data acquisition through to portfolio optimisation.

The analysis is grounded entirely in an original Jupyter notebook containing the underlying calculations, figures and outputs; this report restructures and interprets that analysis in a format suitable for academic and professional review. As a planned extension of this project, the analytical outputs have been prepared for an interactive Streamlit dashboard intended to visualise portfolio performance, risk analytics and optimisation results for a non-technical audience.

1.2 Scope of the Study

The study is confined to a portfolio of 30 U.S.-listed equities drawn from seven sectors, analysed over the five-year period from 1 January 2020 to 1 January 2025 using daily closing price data. The analysis begins with an equally weighted allocation as a neutral baseline and proceeds to portfolio optimisation through Monte Carlo simulation. The scope does not extend to transaction costs, taxation, portfolio rebalancing over time, short selling, out-of-sample validation, or asset classes beyond public equities. All results are derived from historical data and should be interpreted as descriptive of past performance rather than predictive of future returns.

2. Objectives

2.1 Research Aim

The aim of this study is to construct, evaluate and optimise a diversified portfolio of U.S. equities using the principles of Modern Portfolio Theory, and to present the resulting analysis in a manner suitable for academic and professional review.

2.2 Research Objectives

In pursuit of this aim, the study sets out to:

- Download and validate historical stock price data for a diversified set of U.S. equities.
- Calculate daily and cumulative returns for each constituent stock.
- Construct an equally weighted portfolio and compute its expected return and volatility.

- Analyse the covariance and correlation between the returns of the constituent assets.
- Evaluate portfolio performance using risk-adjusted measures, including the Sharpe Ratio, rolling volatility and maximum drawdown.
- Generate a large number of randomly weighted portfolios through Monte Carlo simulation to approximate the Efficient Frontier.
- Identify and interpret the Maximum Sharpe and Minimum Volatility portfolios, and communicate the findings in a structured research report.

3. Literature Review

Modern Portfolio Theory rests on the principle that a rational investor should seek to maximise expected return for a given level of risk, or equivalently minimise risk for a given level of expected return, rather than evaluating assets in isolation. The theory's central contribution is to formalise diversification: because the returns of individual assets are not perfectly correlated, combining assets within a portfolio can reduce overall risk without a proportional reduction in expected return. This principle is operationalised through the covariance and correlation between asset returns, which measure respectively the degree to which two assets move together and the strength and direction of that relationship on a scale bounded between minus one and plus one. Portfolios composed of weakly or negatively correlated assets benefit more from diversification than portfolios composed of highly correlated assets, since offsetting movements dampen aggregate volatility.

A portfolio's risk-adjusted performance is commonly assessed using the Sharpe Ratio, defined as the excess return earned above a risk-free rate per unit of volatility undertaken. A higher Sharpe Ratio indicates that a portfolio generates more return for each unit of risk, and the measure is widely used to compare portfolios with differing risk profiles on a consistent basis. Building on this framework, the Efficient Frontier represents the set of portfolios offering the highest expected return for each level of risk; portfolios lying below the frontier are considered inefficient, since an alternative allocation exists that offers either a higher return for the same risk or the same return for lower risk. Two portfolios on the frontier are of particular interest: the Maximum Sharpe Portfolio, which achieves the highest risk-adjusted return, and the Minimum Volatility Portfolio, which achieves the lowest attainable level of risk.

Because the Efficient Frontier cannot generally be derived analytically for a large number of assets without solving a constrained quadratic optimisation problem, it is commonly approximated using Monte Carlo simulation: a large number of randomly weighted portfolios are generated, their expected return, volatility and Sharpe Ratio computed, and the resulting distribution used to trace the boundary of efficient allocations. This simulation-based approach, applied throughout Section 6.4 of this report, provides a computationally tractable and widely used method of approximating the frontier without requiring a closed-form solution.

A well-documented limitation of the mean-variance framework underlying both the analytical and simulation-based approaches is that its inputs, expected returns, variances and covariances, must themselves be estimated from historical data, and expected returns in particular are known to be estimated with substantially greater error than volatility or correlation. Because portfolio weights derived from mean-variance optimisation are highly sensitive to the expected return assumptions used, small estimation errors can produce disproportionately large shifts in the resulting optimal allocation, a property that has motivated extensions such as the Black-Litterman model, discussed further in Section 7.6. This sensitivity is a central consideration when interpreting the optimisation results presented in this report.

Notwithstanding its theoretical elegance, Modern Portfolio Theory rests on assumptions that do not hold precisely in real markets, including the assumption that asset returns are normally distributed, that correlations remain stable over time, and that investors can transact without cost. These limitations, together with their implications for the present analysis, are considered further in Section 7.

4. Methodology

The analytical framework applies the concepts introduced in Section 3 to the portfolio described in Section 5. Expected portfolio return was calculated as the dot product of the portfolio weight vector and the vector of mean daily returns of the constituent stocks, annualised by multiplying the daily figure by 252, the conventional number of trading days in a calendar year:

$$E[Rp] = (w^T \cdot \mu) \times 252$$

where w is the vector of portfolio weights and μ is the vector of mean daily returns of the constituent stocks. Portfolio volatility was calculated as the square root of the matrix product of the weight vector, the covariance matrix of daily returns, and the transposed weight vector, likewise annualised using the square root of 252:

$$\sigma p = \sqrt{(w^T \Sigma w) \times 252}$$

where Σ is the covariance matrix of daily returns. This formulation accounts not only for the individual volatility of each asset but for the extent to which assets move together, as captured by the covariance matrix. The Sharpe Ratio was calculated as:

$$Sharpe = (E[Rp] - R_f) / \sigma p$$

where R_f is the assumed annual risk-free rate of 4%, representing the approximate yield on short-term U.S. Treasury securities. Rolling volatility was calculated as the annualised standard deviation of portfolio returns within a moving 30-day window, selected because it is responsive to short-term shifts in market conditions while remaining smooth enough to identify sustained periods of elevated risk. Maximum drawdown was calculated as:

$$Drawdown(t) = (Cumulative(t) - RunningMax(t)) / RunningMax(t)$$

tracking the portfolio's cumulative running maximum value and identifying the largest percentage decline from that running peak observed at any point in the sample period.

The Efficient Frontier was approximated through a Monte Carlo simulation of 10,000 randomly weighted portfolios, each constrained to sum to 100% with short selling excluded. For every simulated portfolio the expected annual return, annualised volatility and Sharpe Ratio were computed using the formulae above, and a fixed random seed (`np.random.seed(18)`) was used to ensure that the simulation is reproducible. The portfolio achieving the highest Sharpe Ratio within the simulated set was designated the Maximum Sharpe Portfolio, and the portfolio achieving the lowest volatility was designated the Minimum Volatility Portfolio.

5. Data Collection

5.1 Data Source

Historical daily price data were obtained using the `yfinance` Python library, providing programmatic access to Yahoo Finance market data, for the five-year period from 1 January 2020 to 1 January 2025. Prices were downloaded on an adjusted-close basis (`auto_adjust=True`), which accounts for dividends and stock splits, ensuring that return calculations reflect total investment performance rather than price movements alone.

5.2 Portfolio Selection and Composition

The portfolio was constructed to span seven major sectors of the U.S. economy rather than concentrating on a single industry, reducing concentration risk and providing exposure to distinct economic drivers, including interest rates, consumer spending, commodity prices and business investment. Technology was assigned the largest number of constituents in recognition of its substantial weight within modern U.S. equity markets.

Table 5.1: Portfolio Constituents by Sector

Sector	Constituent Tickers	No. of Stocks
Technology	AAPL, MSFT, NVDA, AMZN, GOOG, META, AMD, ORCL	8
Financials	JPM, BAC, GS, V, MA	5
Healthcare	JNJ, PFE, LLY, MRK, ABBV	5
Consumer	WMT, COST, PG, KO, MCD	5
Automobiles	TSLA, GM, F	3
Energy	XOM, CVX	2
Industrials	CAT, GE	2

This selection yields a total of 30 equities. Amazon and Alphabet are classified under Technology in line with the notebook's sector grouping, reflecting their significant technology and cloud computing operations notwithstanding their formal consumer discretionary or communication services classifications under standard industry taxonomies.

5.3 Data Preprocessing and Portfolio Construction

Following download, the dataset was validated for structure, dimensions and missing values, and daily percentage returns were calculated for each stock using `pct_change()`, with the resulting leading row of undefined values removed via `dropna()`. Working with returns rather than raw prices standardises performance across assets measured on different price scales and forms the basis for all subsequent calculations, including cumulative returns, covariance, correlation, portfolio-level return and volatility, the Sharpe Ratio, and the Monte Carlo simulation.

The baseline portfolio was constructed using an equal-weighting scheme, with each of the 30 constituent stocks assigned an identical weight of $1/30$, approximately 3.33%, verified through the constructed weights DataFrame (`weights_df`) to sum to 100%. An equal-weighted allocation was selected as the starting point because it provides a neutral benchmark that does not presuppose any view on the relative attractiveness of individual securities, allowing differences in portfolio performance to arise solely from the assets' own returns and their interactions with one another.

6. Results

6.1 Correlation and Covariance Analysis

The covariance and correlation matrices of daily returns were computed across all 30 constituents. Owing to its size, the full 30-by-30 matrix is not reproduced in the body of this report and is available within the accompanying notebook.

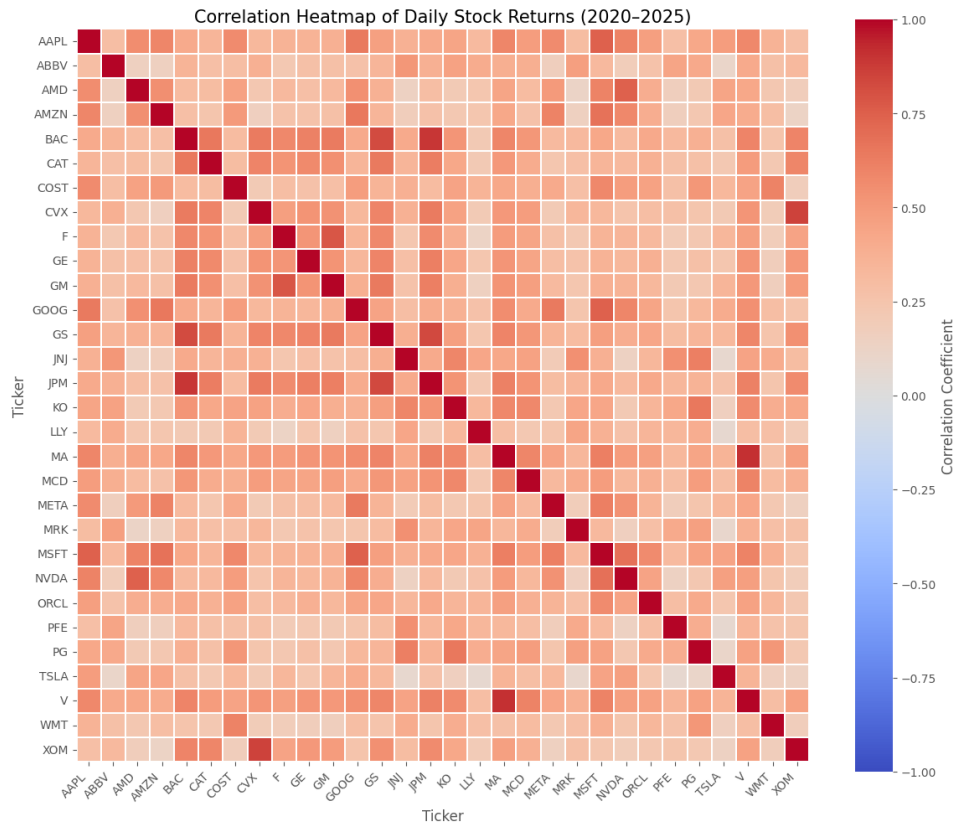


Figure 6.1: Correlation Heatmap of Daily Stock Returns (2020-2025)

Technology holdings, Apple, Microsoft, NVIDIA, Alphabet, Amazon and Advanced Micro Devices, exhibit relatively strong positive correlations, reflecting shared exposure to innovation-driven growth sentiment. Financial institutions, including JPMorgan, Bank of America and Goldman Sachs, are similarly correlated, consistent with their shared sensitivity to interest rates and macroeconomic conditions, as are Chevron and Exxon Mobil through their common exposure to energy prices. Healthcare and consumer staples holdings, such as Johnson & Johnson, Pfizer, Procter & Gamble and Walmart, display comparatively weaker correlations with the technology cluster, and it is this cross-sector variation, rather than the presence of negatively correlated assets, none of which are observed in the dataset, that provides the portfolio's principal source of diversification.

6.2 Return and Growth Analysis

Descriptive statistics of daily returns show that NVIDIA (NVDA) recorded the highest average daily return at 0.3050%, followed by Tesla (TSLA) at 0.2997% and Eli Lilly (LLY) at 0.1663%, while Pfizer (PFE) produced the lowest average daily return at 0.0052%, with Johnson & Johnson and Coca-Cola similarly modest.

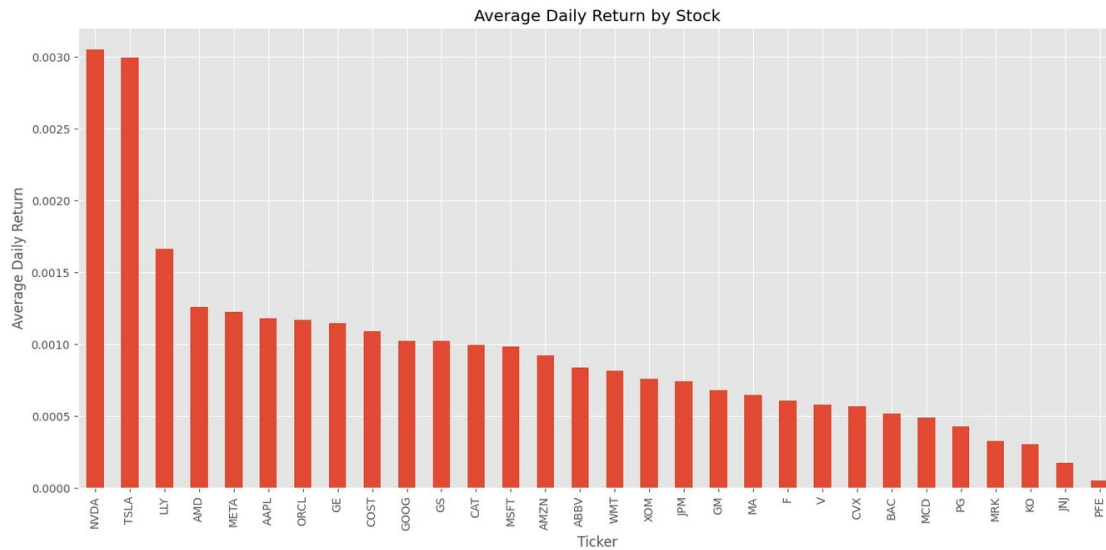


Figure 6.2: Average Daily Return by Stock

Returns are unevenly distributed across the portfolio: growth-oriented technology and healthcare names generated average daily returns considerably above those of the more defensive constituents, foreshadowing the risk-return trade-offs examined in Section 6.3.

Cumulative returns, computed as the running product of one plus the daily return series, reveal that the five strongest long-term performers were NVIDIA, Tesla, Eli Lilly, Apple and Costco.

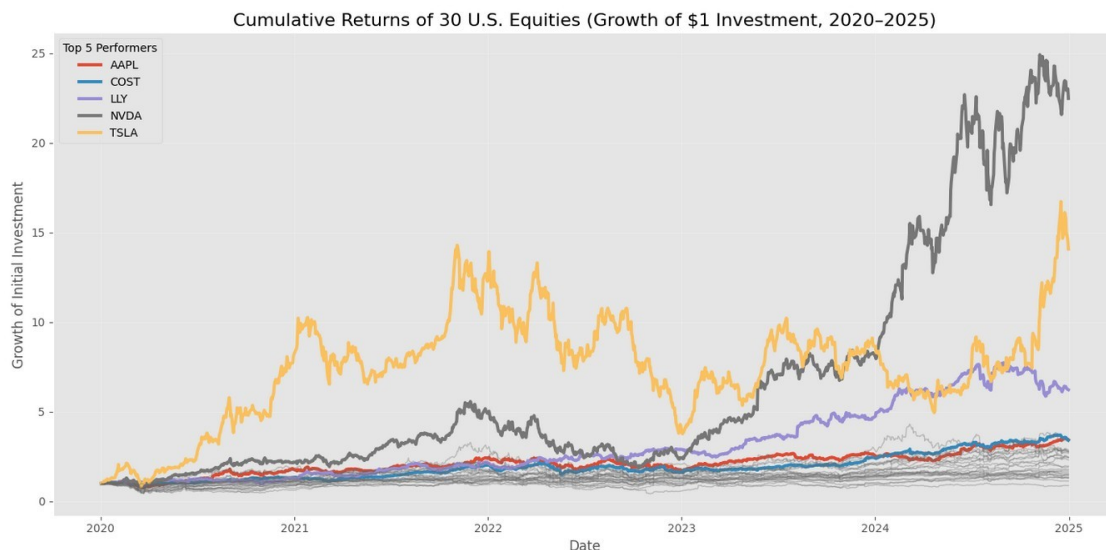


Figure 6.3: Cumulative Returns of 30 U.S. Equities (Growth of a \$1 Investment, 2020-2025)

NVIDIA delivered the highest cumulative return by a considerable margin despite pronounced volatility between 2022 and 2023, while Tesla exhibited the most volatile trajectory among the top performers yet

still produced exceptional long-term growth. Eli Lilly, Apple and Costco achieved comparatively steadier appreciation, illustrating that the highest average daily returns do not necessarily translate into the highest cumulative returns, since compounding depends on the sequence of returns rather than their arithmetic mean alone. Taken together, the strongest long-term gains were generally accompanied by greater price fluctuation, whereas more consistent, lower-volatility growth, as seen in Eli Lilly, Apple and Costco, produced stronger risk-adjusted outcomes.

This divergence between the return rankings in Figure 6.2 and the cumulative growth rankings in Figure 6.3 reflects the well-established relationship between arithmetic and geometric returns: the geometric, compounded return earned by an investor is always less than or equal to the arithmetic average of the underlying periodic returns, with the gap between the two widening as volatility increases, an effect commonly termed volatility drag. A stock with a high average daily return but substantial day-to-day dispersion, as observed for Tesla, therefore surrenders a larger share of its arithmetic performance to compounding than a stock with a lower but steadier average return, such as Costco. This relationship, rather than any change in the underlying data, explains why the ranking of stocks by average daily return in Section 6.2 does not correspond exactly to their ranking by cumulative growth, and reinforces that volatility is a genuine cost to a long-term investor rather than a purely theoretical risk measure.

6.3 Risk Analysis

The equal-weighted portfolio recorded a daily volatility of approximately 1.35%, annualising to 21.40%, against an expected annual return of 23.78% (Section 6.2). This level of risk reflects the portfolio's exposure to higher-volatility growth stocks such as NVIDIA, Tesla and Meta, offset by the diversification benefits identified in Section 6.1, which prevented overall portfolio risk from rising to the level of its most volatile individual holdings.

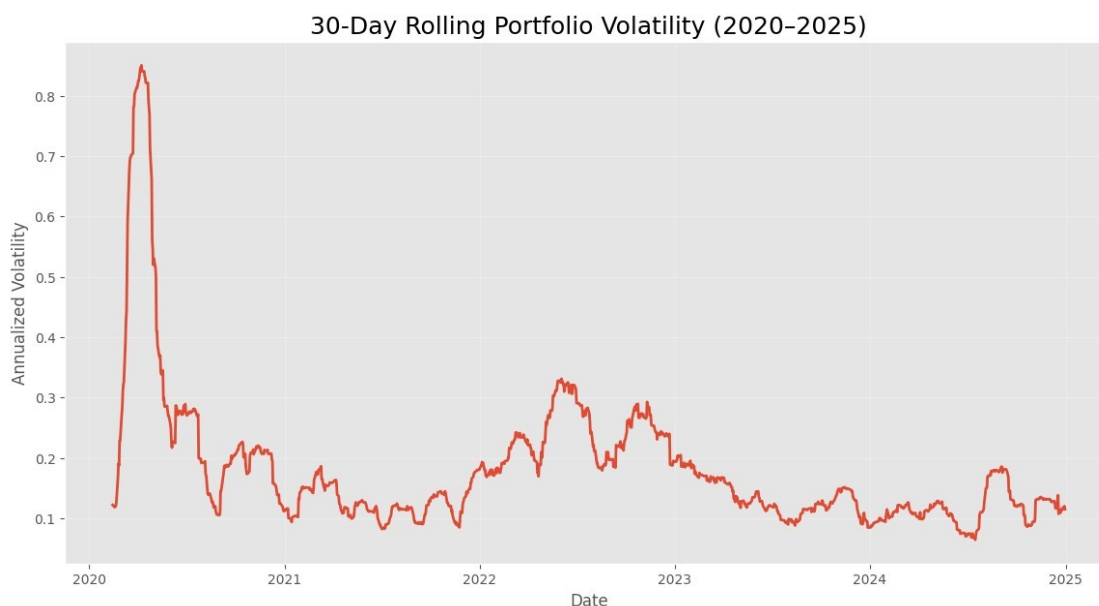


Figure 6.4: 30-Day Rolling Portfolio Volatility (2020-2025)

Rolling volatility averaged 18.10% across the sample period but peaked at 85.03% during the early months of 2020, coinciding with the onset of the COVID-19 pandemic, before declining to a minimum of 6.42% as markets stabilised. A supplementary check using a 90-day rolling window confirmed that this volatility spike remained evident, albeit smoother, indicating that the elevated risk reflected genuine market conditions rather than the specific choice of a 30-day window. This confirms that portfolio risk is dynamic rather than constant, rising sharply during periods of acute market stress and subsiding as conditions normalise.

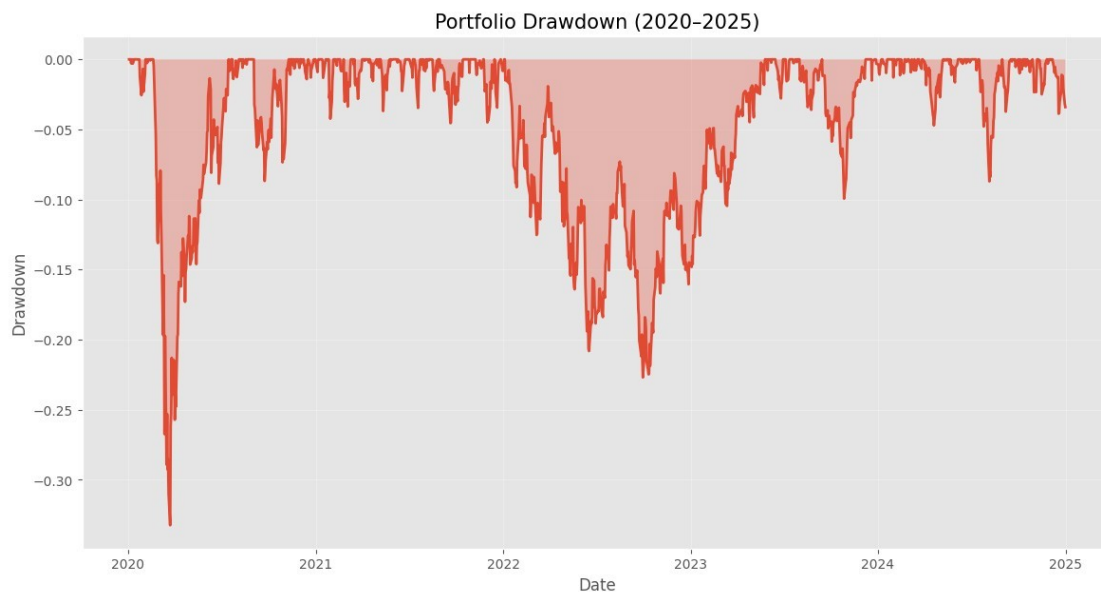


Figure 6.5: Portfolio Drawdown (2020-2025)

The portfolio recorded a maximum drawdown of 33.24%, indicating that an investor entering at the previous peak would have experienced a temporary decline of approximately one-third before recovery. Given that the sample period spans the COVID-19 crash, the 2022 bear market and a period of rapidly rising interest rates, a drawdown of this magnitude is consistent with a portfolio holding a substantial allocation to growth-oriented technology stocks. The drawdown series also shows repeated recoveries to new highs, indicating a degree of resilience despite these episodes of elevated volatility.

The practical significance of a 33.24% drawdown depends heavily on an investor's time horizon and liquidity needs. An investor able to remain invested throughout the decline would, based on the recovery pattern observed in the drawdown series, have recouped the loss within the five-year sample; an investor required to liquidate part of the portfolio during the trough, whether due to a fixed withdrawal schedule or a margin constraint, would crystallise a substantially worse outcome than the annualised return and volatility figures alone would suggest. This is a manifestation of sequence-of-returns risk, whereby the timing of cash flows relative to periods of drawdown, rather than the average return over the full period,

determines the realised outcome for an investor who is not fully buy-and-hold. Maximum drawdown is therefore a materially different risk measure from volatility: whereas volatility summarises the average dispersion of returns, drawdown captures the worst realised outcome for an investor unfortunate enough to require liquidity at the least favourable moment.

With an assumed annual risk-free rate of 4%, the equal-weighted portfolio achieved a Sharpe Ratio of 0.924, indicating that it generated approximately 0.92 units of excess return per unit of risk. This falls just below the commonly cited benchmark of 1.0 for strong risk-adjusted performance, suggesting that while the equal-weight allocation compensated investors adequately for the risk undertaken, it did not represent the most efficient available use of capital, a proposition tested directly through the optimisation exercise in Section 6.4.

6.4 Portfolio Optimisation

A Monte Carlo simulation of 10,000 randomly weighted portfolios was used to approximate the Efficient Frontier and to identify the Maximum Sharpe and Minimum Volatility portfolios described in Section 3.

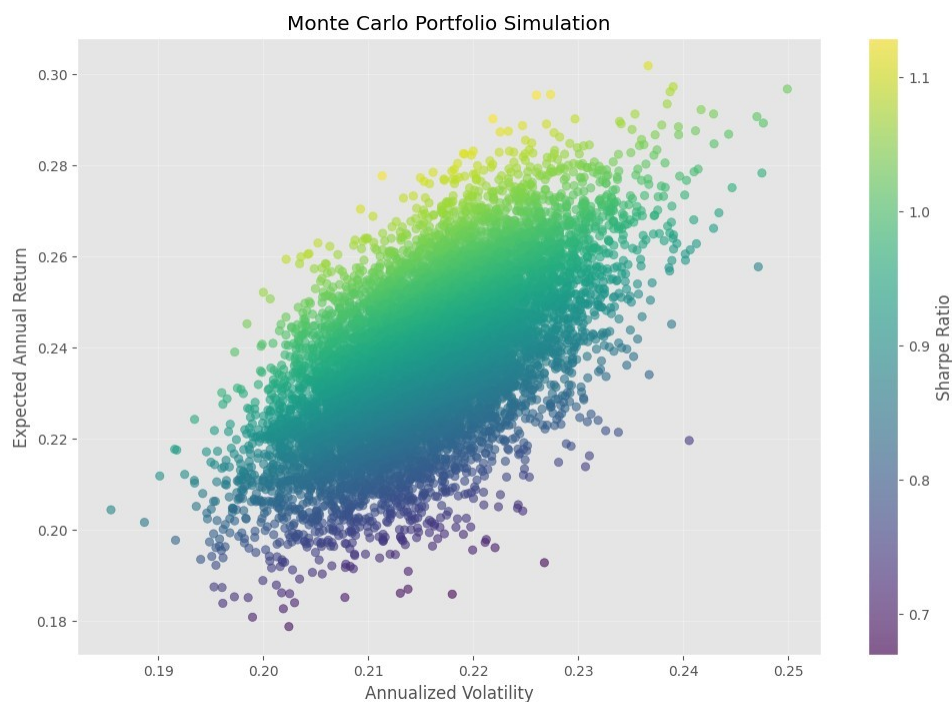


Figure 6.6: Monte Carlo Portfolio Simulation (10,000 Portfolios)

The simulated portfolios trace a clear positive relationship between risk and return, consistent with Modern Portfolio Theory: higher expected returns require accepting greater volatility. Among the 10,000 simulations, the portfolio with the highest Sharpe Ratio achieved an expected annual return of 29.53% at an annualised volatility of 22.60%, a Sharpe Ratio of 1.13, compared with 0.92 for the equal-weight benchmark.

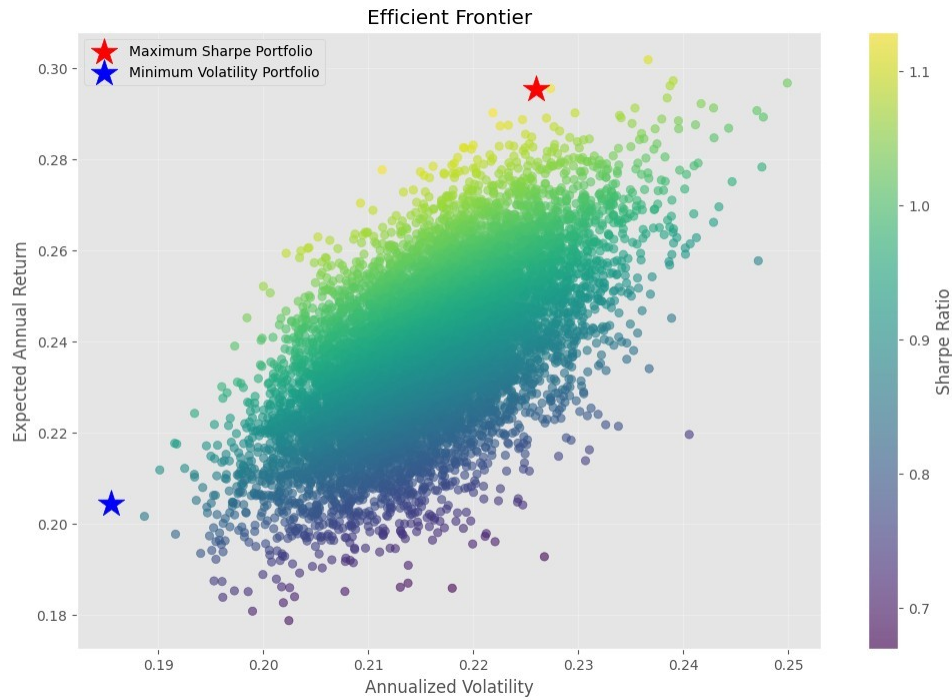


Figure 6.7: Efficient Frontier with Maximum Sharpe and Minimum Volatility Portfolios

Note on the underlying analysis: the notebook's narrative discussion of this figure refers at one point to "100,000 simulated portfolios"; this is inconsistent with the Monte Carlo simulation code, which explicitly sets the number of simulated portfolios to 10,000, a figure stated consistently elsewhere in the notebook. This report follows the value implemented in the code and flags the discrepancy here for the author's attention.

Table 6.1: Optimised Portfolios Compared with the Equal-Weight Benchmark

Portfolio	Expected Return	Annualised Volatility	Sharpe Ratio
Equal-Weight Portfolio	23.78%	21.40%	0.92
Maximum Sharpe Portfolio	29.53%	22.60%	1.13
Minimum Volatility Portfolio	20.44%	18.55%	0.89

The Maximum Sharpe Portfolio improves expected return by 5.75 percentage points relative to the equal-weight benchmark while increasing volatility by only 1.20 percentage points, indicating that the gain stems primarily from a more efficient allocation of capital rather than from a proportionally greater assumption of risk. The Minimum Volatility Portfolio, by contrast, reduces volatility by 2.85 percentage points at the cost of a 3.34 percentage point reduction in expected return, and its Sharpe Ratio of 0.89 falls

marginally below the equal-weight benchmark, indicating that minimising risk in isolation does not necessarily improve risk-adjusted efficiency. The practical implications of this trade-off are considered further in Section 7.

6.5 Portfolio Allocation

The Maximum Sharpe Portfolio does not distribute capital equally; larger weights were assigned to assets that contributed more favourably to the portfolio's risk-adjusted return.

Table 6.2: Largest Allocations in the Maximum Sharpe Portfolio

Rank	Ticker	Company	Portfolio Weight
1	AAPL	Apple Inc.	7.33%
2	XOM	Exxon Mobil Corporation	7.26%
3	COST	Costco Wholesale Corporation	7.13%
4	META	Meta Platforms, Inc.	7.11%
5	AMZN	Amazon.com, Inc.	6.67%

These five holdings collectively account for approximately 35.5% of total capital; the precise weights of the sixth- to tenth-ranked holdings shown in Figure 6.8 are not printed as text within the notebook and are therefore not reproduced numerically here. Readers requiring the complete ranked weight table for all 30 constituents should refer to the `best_weights` output within the accompanying notebook.

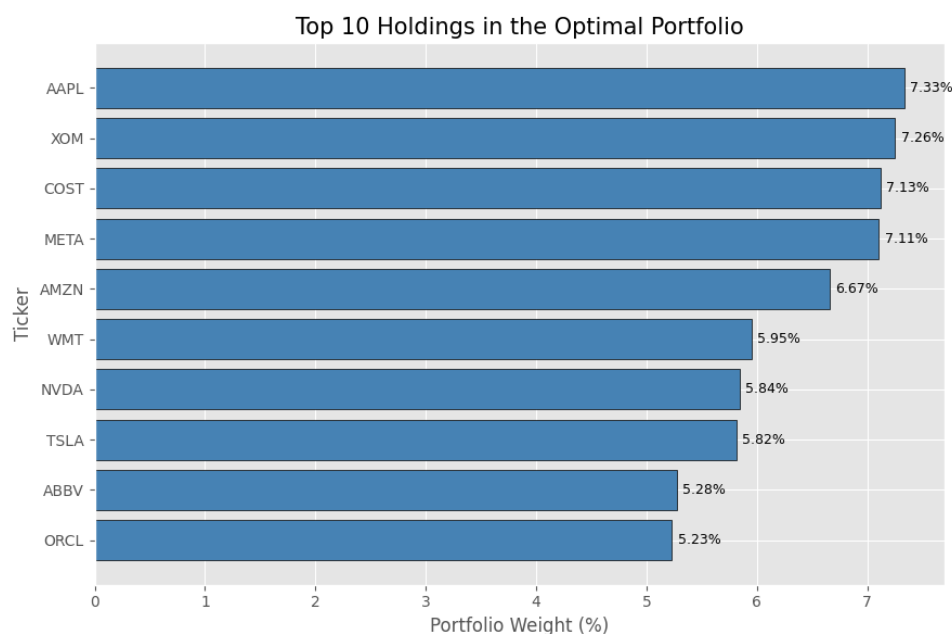


Figure 6.8: Top 10 Holdings in the Optimal Portfolio

The largest allocations span Technology (Apple, Meta, Amazon), Energy (Exxon Mobil) and Consumer (Costco) rather than a single sector, and no individual asset dominates the portfolio. This cross-sector composition indicates that the improved Sharpe Ratio identified in Section 6.4 arises from more efficient weighting of existing diversification benefits, consistent with the correlation patterns discussed in Section 6.1, rather than from concentration in a narrow set of correlated assets. Notably, Exxon Mobil's inclusion among the largest holdings, despite recording one of the more modest average daily returns among the 30 constituents, illustrates that the optimiser rewards assets for their diversification contribution as well as their standalone return; Exxon Mobil's comparatively weak correlation with the technology-dominated core of the portfolio allows it to reduce aggregate volatility in a way that a purely return-ranked selection would not capture.

6.6 Portfolio Improvement Comparison

Table 6.3 draws together the performance of the three portfolios examined in this study, the original equal-weighted allocation and the two portfolios identified through Monte Carlo optimisation, to summarise the overall improvement achieved through reallocation of capital.

Table 6.3: Summary Comparison of Portfolio Strategies

Portfolio	Expected Return	Annualised Volatility	Sharpe Ratio	Change in Sharpe Ratio vs Equal-Weight
Equal-Weight Portfolio	23.78%	21.40%	0.92	(benchmark)
Maximum Sharpe Portfolio	29.53%	22.60%	1.13	+0.21 (+22.8%)
Minimum Volatility Portfolio	20.44%	18.55%	0.89	-0.03 (-3.3%)

The comparison demonstrates that portfolio optimisation improved risk-adjusted performance materially, but not uniformly across objectives. The Maximum Sharpe Portfolio increased the Sharpe Ratio by approximately 23% relative to the equal-weighted benchmark, achieved through reallocation toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon rather than through any change in the underlying 30-stock universe, confirming that capital allocation, not security selection, was the principal driver of the improvement. The Minimum Volatility Portfolio, in contrast, achieved its stated objective of reducing risk, cutting annualised volatility by 2.85 percentage points, but its Sharpe Ratio fell marginally below the equal-weight benchmark, illustrating that minimising volatility in isolation is not equivalent to maximising risk-adjusted return. Together, these results confirm the central proposition tested in this study: that an appropriately weighted portfolio can achieve a superior risk-return trade-off to a naively diversified one without altering the set of assets held, though the specific objective function chosen,

maximising the Sharpe Ratio or minimising volatility, materially shapes the resulting allocation and its practical suitability for different investor risk preferences.

7. Discussion

7.1 Portfolio Performance and Optimisation in Context

The central finding of this study, that reallocating capital from an equal-weighted position toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon raised the portfolio's Sharpe Ratio from 0.92 to 1.13, provides direct empirical support for the proposition at the heart of Modern Portfolio Theory: risk-adjusted performance is driven substantially by how capital is distributed across assets, not solely by which assets are held. The equal-weight portfolio and the Maximum Sharpe Portfolio hold an identical universe of 30 stocks; the entire improvement in risk-adjusted return is therefore attributable to reweighting rather than to security selection, a distinction with direct relevance for portfolio management practice, where naive diversification, spreading capital evenly across a chosen set of assets, is frequently used as a default in the absence of a more considered allocation strategy.

It is instructive that the optimisation process did not simply concentrate capital in the portfolio's highest historical performers. NVIDIA and Tesla, the two strongest individual performers identified in Section 6.2, do not appear among the five largest holdings of the Maximum Sharpe Portfolio in Section 6.5; instead, capital was directed toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon, holdings that combine solid historical returns with more moderate volatility and, in the case of Exxon Mobil, comparatively weak correlation with the technology-dominated core of the portfolio. This outcome illustrates a point that is easily overlooked when interpreting return rankings in isolation: an asset's contribution to portfolio efficiency depends jointly on its own return and volatility and on its covariance with every other holding, not on its standalone performance.

The economic significance of the 0.92-to-1.13 improvement in Sharpe Ratio should not be understated. The modest increase in volatility that accompanied it, reported in Section 6.4, compounds over a multi-year holding period under standard reinvestment assumptions, meaning that an improvement that appears small on an annual basis produces a materially different investment outcome over the five-year study horizon and beyond. This underscores that the reweighting exercise, rather than the selection of the underlying 30-stock universe, was the principal driver of the improvement in risk-adjusted performance.

7.2 The Efficient Frontier and the Risk-Return Trade-off

The comparison between the Maximum Sharpe and Minimum Volatility portfolios in Section 6.4 illustrates that risk minimisation and risk-adjusted return maximisation are not the same objective. An investor targeting the lowest possible volatility would have accepted a marginally lower Sharpe Ratio (0.89) than the equal-weight benchmark (0.92) in exchange for a 2.85 percentage point reduction in volatility, indicating that the Minimum Volatility Portfolio trades away more expected return than is compensated for by the corresponding reduction in risk. This asymmetry is a direct consequence of the

shape of the Efficient Frontier: portfolios positioned toward its lower-risk end typically exhibit a flatter return profile, meaning that further reductions in volatility are purchased at a proportionally higher cost in foregone return, whereas portfolios positioned nearer the Maximum Sharpe point sit at the section of the frontier where additional return can still be obtained relatively cheaply in risk terms.

This has a practical implication for how the frontier should be used in portfolio construction: rather than treating the Maximum Sharpe Portfolio as a universally optimal target, an investor should locate the point on the frontier consistent with their own risk tolerance. For a mandate with an explicit capital-preservation objective, for example a pension fund nearing its payout phase, the Minimum Volatility Portfolio's lower drawdown potential may justify its reduced Sharpe Ratio despite being formally less efficient. For a mandate with a longer horizon and greater tolerance for interim volatility, the Maximum Sharpe Portfolio offers a more attractive combination of return and risk. The Efficient Frontier is therefore best understood not as identifying a single correct portfolio, but as mapping the complete menu of efficient choices from which an allocation consistent with investor-specific constraints can be selected.

7.3 Significance of the Correlation and Diversification Findings

The correlation analysis in Section 6.1 found no negatively correlated pairs among the 30 constituents, meaning that the diversification achieved by the portfolio was one of degree rather than of direction: sector groupings such as technology, financials and energy exhibited materially different levels of positive correlation with one another, but none moved consistently in opposite directions. This is a realistic feature of an equity-only portfolio, since all 30 holdings share exposure to broad market-wide factors, including monetary policy, macroeconomic growth and investor sentiment, regardless of sector. The practical consequence is that the diversification benefit realised in this study, a Sharpe Ratio improvement rather than a wholesale reduction in market exposure, should be understood as a within-asset-class effect; achieving the stronger diversification associated with genuinely offsetting, negatively correlated positions would most plausibly require extending the asset universe beyond U.S. public equities, for example into government bonds or other asset classes whose returns are less tied to equity market sentiment.

The allocation results in Section 6.5 reinforce this interpretation. The five largest holdings in the Maximum Sharpe Portfolio span Technology, Energy and Consumer sectors and together account for only 35.5% of total capital, indicating that the optimisation process retained broad diversification even after reweighting for efficiency; it did not resolve the correlation structure identified in Section 6.1 by concentrating in a small number of assets, but by adjusting the relative weight given to sectors that already provided meaningful diversification value.

7.4 Portfolio Behaviour During the COVID-19 Market Crash

The rolling volatility and drawdown results in Section 6.3 offer further insight into how the portfolio behaved under acute stress. The spike in rolling volatility to 85.03% and the maximum drawdown of 33.24%, both concentrated around the COVID-19 market shock of early 2020, demonstrate that diversification across sectors, while effective in reducing average portfolio risk over the full five-year

sample, did not insulate the portfolio from a systemic, market-wide shock. This is consistent with a well-documented feature of financial markets during periods of crisis: correlations between risk assets tend to rise sharply as investors reduce exposure to risk broadly rather than selectively, a phenomenon sometimes described as diversification failing precisely when it is most needed. Because the correlation matrix in Section 6.1 was estimated over the full five-year sample, it necessarily averages together the elevated correlations of the crisis period with the more typical correlations of calmer periods, meaning that the diversification benefit implied by the full-sample correlation matrix likely overstates the protection available during acute market stress specifically.

The portfolio's subsequent recovery to new cumulative highs, visible in the drawdown series in Figure 6.5, nonetheless indicates that the underlying holdings retained sufficient long-term growth characteristics to more than offset this episode within the five-year window studied. This pattern, a sharp but temporary drawdown followed by sustained recovery, is broadly consistent with the historical behaviour of diversified U.S. equity portfolios following the 2020 shock, and underscores that the maximum drawdown reported in Section 6.3 reflects a peak-to-trough decline at a single point in time rather than a permanent impairment of capital.

7.5 Limitations of Modern Portfolio Theory in Practice

The results of this study are subject to several limitations inherent in the mean-variance framework applied. The framework assumes that historical mean returns, variances and covariances are reliable estimates of their future counterparts; the optimisation results in Section 6.4 rely on the same historical dataset both to identify the optimal allocation and to evaluate its performance, so the reported improvement in Sharpe Ratio is subject to in-sample bias and has not been validated on unseen, out-of-sample data. Expected returns in particular are known to be estimated with considerable error relative to volatility and correlation, meaning that mean-variance optimisation can be disproportionately sensitive to small changes in the return assumptions, a long-recognised weakness of the Markowitz approach used here.

The theory also assumes that correlations between assets are stable over time, an assumption directly challenged by the behaviour documented in Section 7.4, where correlations across risk assets are understood to increase during periods of crisis. Furthermore, the analysis excludes transaction costs, taxation and the practical constraints of rebalancing a portfolio to match optimised weights on an ongoing basis, all of which would erode the realised advantage of the Maximum Sharpe Portfolio over the equal-weight benchmark in practice; a portfolio that is optimal at a single point in time will typically drift away from that optimum as prices move, requiring rebalancing that itself incurs cost. The risk-free rate was also held constant at 4% throughout the study period, whereas short-term interest rates varied considerably between 2020 and 2025, meaning that the Sharpe Ratios reported in this study represent an approximation rather than a period-accurate risk-adjusted return.

7.6 Assumptions and Directions for Future Work

Several extensions would address the limitations identified above and would represent natural next steps for this project. Incorporating transaction costs and a defined rebalancing schedule, for example quarterly rebalancing back to target weights, would produce a more realistic estimate of achievable, rather than theoretical, risk-adjusted return. Out-of-sample or rolling-window validation, in which the optimal weights are estimated on one period and tested on a subsequent, unseen period, would provide a more rigorous test of whether the Monte Carlo optimisation genuinely improves forward-looking performance or merely fits historical noise, directly addressing the in-sample bias discussed in Section 7.5.

The Black-Litterman model, which blends the market-implied equilibrium return with an investor's own views, offers a means of mitigating the sensitivity of mean-variance optimisation to estimation error in expected returns identified above, and would allow the analysis to incorporate a degree of forward-looking judgement rather than relying exclusively on historical averages. Extending the analysis through factor investing, decomposing returns into exposures to established risk factors such as size, value and momentum, would offer a more granular explanation of why certain holdings, including Apple, Exxon Mobil and Costco, contributed disproportionately to the optimised portfolio's efficiency, and would help distinguish genuine diversification benefit from exposure to a small number of common underlying risk factors. Finally, extending the asset universe beyond U.S. public equities, for example to include government bonds, would test whether the stronger, direction-based diversification discussed in Section 7.3 is achievable in a way that the present equity-only universe does not permit.

7.7 Practical Implications for Investors

Taken together, the findings translate into several practical considerations for portfolio management. Periodic re-optimisation is a worthwhile discipline rather than a one-off exercise, since even a systematic reweighting confined to the same 30-stock universe materially improved risk-adjusted return. The choice between a Maximum Sharpe and a Minimum Volatility orientation should follow explicitly from an investor's objective and time horizon rather than a general preference for one over the other, given that Section 7.2 shows the two portfolios serve materially different purposes. The concentration of the largest optimised weights in Apple, Exxon Mobil, Costco, Meta Platforms and Amazon, none of which was the single highest-returning stock in the sample, further illustrates that a disciplined, correlation-aware allocation process can identify efficient holdings that a return-ranking heuristic alone would overlook.

Finally, the divergence between the portfolio's full-sample correlation structure and its behaviour during the COVID-19 shock, discussed in Section 7.4, is a caution against relying on long-run historical correlation and volatility estimates as a guide to diversification during periods of acute market stress specifically. Investors and analysts using a similar methodology should treat the resulting optimal weights as a reasonable long-run allocation guide rather than a guarantee of protection during a systemic shock, and should consider supplementing historical optimisation with stress-testing against past crisis periods.

8. Conclusion

This report has applied Modern Portfolio Theory to a diversified portfolio of 30 U.S. equities across seven sectors, progressing from data acquisition and descriptive analysis through correlation and covariance analysis, portfolio-level risk and return measurement, and Monte Carlo-based optimisation, meeting each of the research objectives set out in Section 2.2.

The principal finding is that the equal-weighted portfolio, while delivering a historically attractive expected annual return of 23.78% at an annualised volatility of 21.40%, was not the most efficient available allocation of capital. Reweighting toward Apple, Exxon Mobil, Costco, Meta Platforms and Amazon raised the Sharpe Ratio from 0.92 to 1.13 without a proportional increase in risk, while a Minimum Volatility Portfolio demonstrated that risk reduction and risk-adjusted efficiency are distinct objectives that do not necessarily move together. The correlation analysis further established that the portfolio's diversification benefit derived from combining assets across sectors with varying degrees of positive correlation, rather than from negatively correlated holdings, and that this diversification, while reducing average risk, did not prevent a substantial, systemic drawdown during the COVID-19 market shock.

The principal contribution of this project lies in its demonstration of a complete, reproducible portfolio analysis and optimisation pipeline, implemented from first principles in Python and evaluated against the theoretical expectations of Modern Portfolio Theory, together with an honest accounting of where the underlying assumptions of that theory diverge from real-market conditions. As a planned extension, the analytical outputs generated throughout this project have been prepared for an interactive Streamlit dashboard, allowing the performance, risk and optimisation results presented here to be explored dynamically by a non-technical audience.

In summary, this study demonstrates the successful practical application of Modern Portfolio Theory to a real-world equity universe: a rigorous quantitative pipeline, spanning descriptive statistics, covariance and correlation analysis, risk measurement and simulation-based optimisation, was used to show that disciplined capital allocation, rather than security selection alone, is capable of delivering a materially superior risk-adjusted outcome. The resulting Maximum Sharpe and Minimum Volatility portfolios offer investors a concrete, evidence-based choice between two well-defined risk postures, while the project's transparent treatment of its own limitations provides a credible foundation for extension toward out-of-sample validation, factor-based and Black-Litterman approaches, and multi-asset-class diversification. Taken as a whole, the analysis stands as a robust demonstration of quantitative portfolio management capability, combining technical rigour with clear, decision-relevant financial insight, and one that is readily extensible to more sophisticated investment mandates in future work.

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Appendix

A.1 Project Workflow

The analytical workflow followed in the underlying notebook proceeds through the following sequential stages:

- Download historical prices
- Data validation
- Return calculations
- Portfolio statistics
- Risk analysis
- Monte Carlo simulation
- Efficient Frontier
- Report

A.2 Tools and Libraries

- yfinance: download of historical market data
- pandas: data manipulation and analysis
- NumPy: numerical computing and matrix operations
- Matplotlib: static data visualisation (Figures 6.2-6.6, 6.8)
- Seaborn: statistical visualisation, including the correlation heatmap (Figure 6.1)
- Streamlit: planned interactive dashboard for presenting the analysis

A.3 Summary of Key Metrics

Metric	Equal-Weight Portfolio	Maximum Sharpe Portfolio	Minimum Volatility Portfolio
Expected Annual Return	23.78%	29.53%	20.44%
Annualised Volatility	21.40%	22.60%	18.55%
Sharpe Ratio	0.92	1.13	0.89
Maximum Drawdown	33.24%	-	-
Average Rolling Volatility (30-day)	18.10%	-	-

Note: the notebook computes maximum drawdown and rolling volatility only for the equal-weighted portfolio; these measures were not recalculated for the Maximum Sharpe or Minimum Volatility portfolios within the analysed notebook. This is noted here rather than estimated, in keeping with the requirement not to invent results beyond those contained in the source material.

A.4 Assumptions

- Study period: 1 January 2020 to 1 January 2025 (five years of daily data).
- Annualisation convention: 252 trading days per year.
- Risk-free rate: 4% per annum, held constant throughout the study period.
- Monte Carlo simulation size: 10,000 randomly weighted portfolios, generated with a fixed random seed (`np.random.seed(18)`) to ensure reproducibility.
- Rolling volatility window: 30 trading days.

A.5 Author Information

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Project: Quant Foundations – Project 1

This report was prepared from the author's original Jupyter Notebook analysis. All calculations, visualisations, figures, tables, and results presented herein were produced from the underlying analysis and represent the author's original work.